



SC2207 Lab 5 Report

SDAB Team 4

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5. Nelson Wong Chong Yee
6. Wong Ye Fei

APPENDIX C: INDIVIDUAL CONTRIBUTION FORM

Full Name	Individual Contribution to Lab 1 Submission	Percentage of Contribution	Signature
NELSON WONG CHONG YEE	Organize the team to work together List the entity sets and relationships as a draft before drawing the ER diagram. check for errors in the ER diagram.	16.67%	
BUI GIA NHAT MINH	Analyse the requirements of the lab and description of the program to come up with entity sets and relations. Draft one of the ER diagrams to cross check with others Help in check for errors and feedback for other diagrams	16.67%	
NGUYEN THU HUYEN	- Draw a version of ER diagram. - Check for errors in ER diagrams to choose the best one. - Finalize diagram.	16.67%	
FOO JIN HAO	Draft a version of ER diagram Provide feedback and suggestions for other diagrams Check for minor error in the ER diagram	16.67%	
WONG YE FEI	Draft a version of ER diagram for reference Giving suggestions for other diagrams Check for errors in ER diagrams	16.66%	
PHAM GIA PHUC	Summarize errors in ER diagrams after feedback meeting Make suggestion for summarized feedback Fix those errors to finalize the ER diagram	16.66%	

APPENDIX D: USE OF AI TOOL(S) IN LAB WORK

Each team member should indicate either A or B.

A. I affirm that my contribution(s) to the lab work is my own, produced without help from any AI tool(s)

B. I affirm that my contribution(s) to the lab work has been produced with the help from AI tool(s)

Full Name	Signature	Date	A or B
Bui Gia Nhat Minh		29 th Aug 2025	A
Pham Gia Phuc		29 th Aug 2025	A
Nguyen Thu Huyen		29 th Aug 2025	A
Foo Jin Hao		29 th Aug 2025	A
Nelson Wong Chong Yee		29 th Aug 2025	A
Wong Ye Fei	wong	29 th Aug 2025	A

By signing this form, you declare that the above affirmation made is true and that you have read and understood NTU's policy on the use of AI tools.

If any team member answered B, the team member(s) must indicate and replicate the table below for every instance AI tool(s) is used.

Name of AI tool	< For example, ChatGPT >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	
Output generated	< Insert the response verbatim from ChatGPT >
Output screenshot	
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >

In this lab, we will create tables, insert data, and solves required queries using SQL statements. Here are the assumptions made for our tables:

- Each vehicle has its own IU_OBU_ID that can uniquely determine the vehicle registration number, but it is still necessary for parking
- Parking Record must record one and exactly one UID to determine the actual amount paid for the parking session, either coming from the transaction of short-term parking, or coming from the seasonal pass, but not from both.
- The calculation of the amount payable for parking sessions is handled by the application layer rather than the database. Therefore, the database will only store the amount as a string or text value, without performing any computational logic.

Our lab report consists of 2 parts:

Part 1: Table Initialization (Page 5 – 22)

Part 2: Solving queries in Appendix B (Page 22 – 28)

--Tables Initialization -----

```
CREATE TABLE NormalCarpark(  
ID CHAR(6),  
free_parking_period VARCHAR(100) ,  
night_parking INT CHECK (night_parking IN (0,1)),
```

```
PRIMARY KEY (ID)  
);
```

```
INSERT INTO NormalCarpark( ID, free_parking_period, night_parking)
```

```
VALUES
```

```
('HDB001', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB002', 'Sunday & Public Holiday, 7:00am–10:30pm', 0),  
( 'HDB003', 'Public Holiday only, 7:00am–10:30pm', 1),  
( 'HDB004', 'Sunday only, 7:00am–10:30pm', 1),  
( 'HDB005', 'No free parking', 0),  
( 'HDB006', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB007', 'Sunday only, 7:00am–10:30pm', 0),  
( 'HDB008', 'Public Holiday only, 7:00am–10:30pm', 1),  
( 'HDB009', 'No free parking', 0),  
( 'HDB010', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB011', 'Sunday only, 7:00am–10:30pm', 1),  
( 'HDB012', 'Public Holiday only, 7:00am–10:30pm', 0),  
( 'HDB013', 'No free parking', 0),  
( 'HDB014', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB015', 'Sunday only, 7:00am–10:30pm', 0),  
( 'HDB016', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB017', 'Public Holiday only, 7:00am–10:30pm', 0),  
( 'HDB018', 'No free parking', 0),  
( 'HDB019', 'Sunday & Public Holiday, 7:00am–10:30pm', 1),  
( 'HDB020', 'Sunday only, 7:00am–10:30pm', 0);
```

```
-----  
CREATE TABLE MSCP(  
ID CHAR(6),  
clearance_height FLOAT,  
number_of_decks INT,
```

```
PRIMARY KEY (ID),
```

```
FOREIGN KEY (ID) REFERENCES NormalCarpark(ID) ON DELETE CASCADE ON UPDATE  
CASCADE
```

);

INSERT INTO MSCP(ID, clearance_height, number_of_decks)

VALUES

('HDB001', 2.2, 6),
('HDB005', 2.2, 8),
('HDB006', 2.4, 10),
('HDB007', 2.1, 5),
('HDB008', 2.3, 9),
('HDB009', 2.0, 3),
('HDB011', 2.1, 4),
('HDB014', 2.2, 7),
('HDB018', 2.4, 11),
('HDB019', 2.2, 8);

CREATE TABLE ParkingLots(

ID CHAR(6),
color VARCHAR(20),
number_of_lots INT,

PRIMARY KEY(ID,color),

FOREIGN KEY(ID) REFERENCES NormalCarpark(ID) ON DELETE CASCADE ON UPDATE
CASCADE

);

INSERT INTO ParkingLots(ID,color ,number_of_lots)

VALUES

('HDB001', 'Red', 80),
('HDB001', 'Yellow', 40),
('HDB001', 'White', 10),
('HDB002', 'Red', 60),
('HDB002', 'White', 20),
('HDB003', 'Red', 35),
('HDB004', 'Red', 70),
('HDB004', 'Yellow', 10),
('HDB005', 'Red', 50),
('HDB005', 'White', 15),
('HDB006', 'Red', 75),
('HDB006', 'White', 25),
('HDB006', 'Bi-color', 8),

```
('HDB007', 'Red', 55),
('HDB007', 'White', 20),
('HDB008', 'Red', 90),
('HDB009', 'Red', 40),
('HDB009', 'Yellow', 5),
('HDB010', 'Red', 13),
('HDB010', 'White', 12),
('HDB011', 'Red', 65),
('HDB012', 'Red', 70),
('HDB013', 'Red', 60),
('HDB014', 'Red', 80),
('HDB015', 'Red', 75),
('HDB016', 'Red', 90),
('HDB017', 'Red', 85),
('HDB018', 'Red', 95),
('HDB019', 'Red', 60),
('HDB020', 'Red', 55);
```

```
-----
CREATE TABLE BlockAddress(
postal_code INT,
hdb_block_number VARCHAR(10) ,
street_name VARCHAR(30),
carpark_id CHAR(6),

PRIMARY KEY (postal_code),

FOREIGN KEY (carpark_id) REFERENCES NormalCarpark(ID) ON DELETE SET NULL ON
UPDATE CASCADE
);
```

```
INSERT INTO BlockAddress( postal_code, hdb_block_number, street_name,
carpark_id)
VALUES
(560123, '123', 'Ang Mo Kio Ave 3', 'HDB001'),
(560124, '124', 'Ang Mo Kio Ave 3', 'HDB002'),
(310125, '25A', 'Toa Payoh Lor 2', 'HDB003'),
(310126, '26B', 'Toa Payoh Lor 6', 'HDB004'),
(460101, '101', 'Bedok North St 2', 'HDB005'),
(460102, '102', 'Bedok North St 3', 'HDB006'),
(520341, '341', 'Tampines St 33', 'HDB007'),
(520342, '342', 'Tampines St 34', 'HDB008'),
```

(640601, '601', 'Jurong West St 62', 'HDB009'),
(640602, '602', 'Jurong West St 61', 'HDB010'),
(650231, '231', 'Bukit Batok East Ave 5', 'HDB011'),
(650232, '232', 'Bukit Batok St 21', 'HDB012'),
(730401, '401', 'Woodlands Ave 3', 'HDB013'),
(730402, '402', 'Woodlands St 41', 'HDB014'),
(820201, '201', 'Hougang St 21', 'HDB015'),
(820202, '202', 'Hougang Ave 8', 'HDB016'),
(600501, '501', 'Jurong West St 52', 'HDB017'),
(600502, '502', 'Jurong West Ave 1', 'HDB018'),
(680301, '301', 'Choa Chu Kang Ave 4', 'HDB019'),
(680302, '302', 'Choa Chu Kang Loop', 'HDB020'),
(640603, '603', 'Jurong West St 61', 'HDB010'),
(640604, '604', 'Jurong West St 61', 'HDB010'),
(640605, '605', 'Jurong West St 61', 'HDB010'),
(640606, '606', 'Jurong West St 61', 'HDB010'),
(640607, '607', 'Jurong West St 61', 'HDB010'),
(640608, '608', 'Jurong West St 62', 'HDB010'),
(640609, '609', 'Jurong West St 61', 'HDB010'),
(640610, '610', 'Jurong West St 62', 'HDB010'),
(640611, '611', 'Jurong West St 61', 'HDB010');

CREATE TABLE ResidentUnit(
postal_code INT,
unit_number VARCHAR(10),

PRIMARY KEY (postal_code, unit_number),

FOREIGN KEY (postal_code) REFERENCES BlockAddress(postal_code) ON DELETE
CASCADE ON UPDATE CASCADE
);

INSERT INTO ResidentUnit(postal_code, unit_number)
VALUES
(560123, '#01-101'),
(560124, '#02-102'),
(310125, '#03-103'),
(310126, '#04-201'),
(460101, '#05-202'),
(460102, '#11-111'),
(520341, '#07-302'),

(520342, '#08-303'),
(640601, '#09-304'),
(640602, '#10-401'),
(650231, '#11-402'),
(650232, '#12-403'),
(730401, '#13-404'),
(730402, '#20-701'),
(820201, '#16-503'),
(820202, '#18-601'),
(600501, '#14-511'),
(600502, '#10-481'),
(680301, '#06-301'),
(680302, '#07-512'),
(640604, '#03-312'),
(640604, '#04-234'),
(640605, '#09-293'),
(640607, '#01-181'),
(640607, '#20-325'),
(640603, '#10-398'),
(640606, '#09-415'),
(640606, '#16-428'),
(640605, '#13-422'),
(640606, '#15-180'),
(640606, '#16-339'),
(640605, '#11-148'),
(640606, '#06-210'),
(640603, '#10-246'),
(640607, '#04-404'),
(640604, '#20-323'),
(640607, '#17-367'),
(640605, '#08-146'),
(640603, '#20-158'),
(640605, '#01-419');

CREATE TABLE Owner(
NRIC CHAR(9),
name VARCHAR(50),
postal_code INT,
unit_number VARCHAR(10),

PRIMARY KEY (NRIC),

```
FOREIGN KEY (postal_code, unit_number) REFERENCES ResidentUnit(postal_code,
unit_number) ON DELETE SET NULL ON UPDATE CASCADE
);
```

```
INSERT INTO Owner (NRIC, name, postal_code, unit_number) VALUES
```

```
('M0000001A','John', 560123, '#01-101'),
('M0000002B','Peter', 560124, '#02-102'),
('M0000003C','Jack', 310125, '#03-103'),
('M0000004D','Michael', 310126, '#04-201'),
('M0000005E','Fury', 460101, '#05-202'),
('M0000006F','John', 460102, '#11-111'),
('M0000007G','Jack', 520341, '#07-302'),
('M0000008H','Jack', 520342, '#08-303'),
('M0000009I','Angeline', 640601, '#09-304'),
('M0000010J','Sophie', 640602, '#10-401'),
('M0000011K','Michael',640603,'#20-158'),
('M0000012L','Henry',640603,'#20-158'),
('M0000013M','Oslo',640603,'#20-158'),
('F0000001A','Sophie', 650231, '#11-402'),
('F0000002B','Angeline', 730401, '#13-404'),
('F0000003C','Aurelia', 730402, '#20-701'),
('F0000004D','Minnie', 820201, '#16-503'),
('F0000005E','Aster', 820202, '#18-601'),
('F0000006F','Minnie', 600501, '#14-511'),
('F0000007G','Michael', 600502, '#10-481'),
('F0000008H','Sophie', 680301, '#06-301'),
('F0000009I','Harry', 680302, '#07-512'),
('F0000010J','Minnie', 650232, '#12-403'),
('F0000011K','Maria',640605,'#13-422'),
('F0000012L','Norla',640606,'#15-180');
```

```
-----
CREATE TABLE Email(
email VARCHAR(50),
NRIC CHAR(9),
```

```
PRIMARY KEY (email),
```

```
FOREIGN KEY (NRIC) REFERENCES Owner(NRIC)
);
```

```
INSERT INTO Email (email, NRIC) VALUES
('john001@gmail.com', 'M0000001A'),
('e0129780@nus.edu', 'M0000001A'),
('john123@ntu.edu', 'M0000001A'),
('sophie369@gmail.com', 'F0000001A'),
('soso@ntu.edu', 'F0000001A'),
('jackiechan@gmail.com', 'M0000007G'),
('angelina@nus.edu', 'M0000009I'),
('harryoso@ntu.edu', 'F0000009I'),
('mailingaddress@gmail.com', 'M0000010J');
```

```
CREATE TABLE ContactNumber(
contact_number CHAR(8),
NRIC CHAR(9),

PRIMARY KEY (contact_number),

FOREIGN KEY (NRIC) REFERENCES Owner(NRIC)
);
```

```
INSERT INTO ContactNumber (contact_number, NRIC) VALUES
('00000001', 'M0000001A'),
('01234567', 'M0000002B'),
('24681357', 'M0000002B'),
('76531249', 'M0000002B'),
('14714714', 'F0000003C'),
('78945612', 'F0000004D'),
('13052413', 'F0000003C'),
('87469312', 'F0000009I'),
('00000002', 'M0000006F');
```

```
CREATE TABLE VehicleModel(
model VARCHAR(15),
vehicle_type VARCHAR(15),
brand VARCHAR(15),
```

```
PRIMARY KEY (model)
);
```

```
INSERT INTO VehicleModel (model, vehicle_type, brand)
```

VALUES

```
('Car001', 'Car', 'Honda'),
('Car002', 'Car', 'Toyota'),
('Car003', 'Car', 'Tesla'),
('Car004', 'Car', 'Kia'),
('Motor001', 'Motorcycle', 'Honda'),
('Motor002', 'Motorcycle', 'Yamaha'),
('Motor003', 'Motorcycle', 'Honda'),
('Comm001', 'Commercial', 'Toyota'),
('Comm002', 'Commercial', 'Nissan'),
('Comm003', 'Commercial', 'Mitsubishi');
```

CREATE TABLE Vehicle1 (

```
VRN CHAR(6),
IU_OBU_ID CHAR(8) NOT NULL,
color VARCHAR(15),
year_manufactured INT,
model VARCHAR(15),
NRIC CHAR(9),
```

```
PRIMARY KEY (VRN),
UNIQUE (IU_OBU_ID),
```

```
FOREIGN KEY (model) REFERENCES VehicleModel(model) ON UPDATE CASCADE,
FOREIGN KEY (NRIC) REFERENCES Owner(NRIC)
);
```

INSERT INTO Vehicle1 (VRN, IU_OBU_ID, color, year_manufactured, model, NRIC)

VALUES

```
('VRN001','IU000001','white',2018,'Car001','M0000006F'),
('VRN002','OBU00001','red',2019,'Car002','M0000005E'),
('VRN003','OBU00002','yellow',2025,'Car004','F0000009I'),
('VRN004','IU000002','blue',2024,'Car002','M0000003C'),
('VRN005','IU000003','green',2022,'Car003','M0000005E'),
('VRN006','OBU00003','green',2019,'Car003','F0000005E'),
('VRN007','IU000004','pink',2015,'Motor003','M0000003C'),
('VRN008','IU000005','silver',2020,'Car001','M0000001A'),
('VRN009','OBU00004','black',2023,'Car004','F0000002B'),
('VRN010','IU000006','white',2017,'Car002','M0000002B'),
('VRN011','OBU00005','gray',2019,'Motor001','F0000003C'),
('VRN012','IU000007','blue',2021,'Motor002','M0000004D'),
```

```
('VRN013','OBU00006','orange',2024,'Motor003','F0000004D'),
('VRN014','IU000008','red',2022,'Comm001','M0000006F'),
('VRN015','OBU00007','green',2023,'Comm002','F0000006F'),
('VRN016','IU000009','white',2016,'Comm003','M0000007G'),
('VRN017','OBU00008','black',2018,'Car003','F0000008H'),
('VRN018','IU000010','blue',2019,'Car001','M0000008H'),
('VRN019','OBU00009','gray',2020,'Motor002','F0000009I'),
('VRN020','IU000011','yellow',2024,'Car004','M0000010J'),
('VRN021','IU000012','white',2019,'Comm003','M0000011K'),
('VRN022','OBU00010','red',2018,'Car003','F0000012L'),
('VRN023','IU000013','blue',2019,'Car001','M0000013M'),
('VRN024','OBU00011','green',2020,'Car002','F0000010J'),
('VRN025','IU000014','pink',2024,'Car004','M0000012L');
```

```
-----
CREATE TABLE ParkingModeRecord (
  UID VARCHAR(10) NOT NULL,
  parkingmode_ref VARCHAR(30),
  PRIMARY KEY (UID),
);
```

```
INSERT INTO ParkingModeRecord(UID, parkingmode_ref) VALUES
('UID0001','Short-term'),
('UID0002','Short-term'),
('UID0003','Short-term'),
('UID0004','Short-term'),
('UID0005','Short-term'),
('UID0006','Short-term'),
('UID0007','Short-term'),
('UID0008','Short-term'),
('UID0009','Short-term'),
('UID0010','Short-term'),
('UID0011','Short-term'),
('UID0012','Short-term'),
('UID0013','Short-term'),
('UID0014','Short-term'),
('UID0015','Short-term'),
('UID0016','Short-term'),
('UID0017','Short-term'),
('UID0018','Short-term'),
('UID0019','Short-term'),
('UID0020','Short-term'),
```

('UID0021','Short-term'),
('UID0022','Short-term'),
('UID0023','Short-term'),
('UID1001','Seasonal'),
('UID1002','Seasonal'),
('UID1003','Seasonal'),
('UID1004','Seasonal'),
('UID1005','Seasonal'),
('UID1006','Seasonal'),
('UID1007','Seasonal'),
('UID1008','Seasonal'),
('UID1009','Seasonal'),
('UID1010','Seasonal'),
('UID1011','Seasonal'),
('UID1012','Seasonal'),
('UID1013','Seasonal'),
('UID1014','Seasonal'),
('UID1015','Seasonal'),
('UID1016','Seasonal'),
('UID1017','Seasonal'),
('UID1018','Seasonal'),
('UID1019','Seasonal'),
('UID1020','Seasonal'),
('UID1021','Seasonal'),
('UID1022','Seasonal'),
('UID1023','Seasonal'),
('UID1024','Seasonal'),
('UID1025','Seasonal'),
('UID1026','Seasonal'),
('UID1027','Seasonal'),
('UID1028','Seasonal'),
('UID1029','Seasonal'),
('UID1030','Seasonal'),
('UID1031','Seasonal'),
('UID1505','Seasonal'),
('UID1506','Seasonal'),
('UID1507','Seasonal'),
('UID1508','Seasonal');

CREATE TABLE R_SeasonPass_2 (
vehicle_type VARCHAR(30) NOT NULL,

```
carpark_type VARCHAR(30) NOT NULL,  
monthly_rate FLOAT NOT NULL CHECK(monthly_rate>=0),
```

```
PRIMARY KEY (vehicle_type, carpark_type)  
);
```

```
INSERT INTO R_SeasonPass_2 (vehicle_type, carpark_type, monthly_rate) VALUES  
( 'Car', 'MSCP', 100.00),  
( 'Car', 'Surface', 80.00),  
( 'Motorcycle', 'MSCP', 50.00),  
( 'Motorcycle', 'Surface', 30.00),  
( 'Commercial', 'MSCP', 190.00),  
( 'Commercial', 'Surface', 170.00);  
-----
```

```
CREATE TABLE R_Transaction_SeasonPass_1(  
UTR VARCHAR(30),  
date_time DATETIME NOT NULL,  
amount_paid FLOAT NOT NULL CHECK(amount_paid>=0),  
expiry_date DATETIME NOT NULL,  
CHECK(expiry_date>=date_time),  
VRN CHAR(6) ,  
UID VARCHAR(10),  
carpark_ID CHAR(6),  
PRIMARY KEY (UTR),
```

```
FOREIGN KEY(VRN) REFERENCES Vehicle1(VRN)  
ON DELETE CASCADE,
```

```
FOREIGN KEY (carpark_ID) REFERENCES NormalCarpark(ID)  
ON DELETE CASCADE,
```

```
FOREIGN KEY (UID) REFERENCES ParkingModeRecord(UID)  
ON DELETE CASCADE  
);
```

```
INSERT INTO R_Transaction_SeasonPass_1 (UTR, date_time, amount_paid,  
expiry_date, VRN, UID, carpark_ID) VALUES  
( 'UTR001', '2025-01-01', 100.00, '2025-02-01', 'VRN001', 'UID1001', 'HDB006'),  
( 'UTR002', '2025-01-01', 100.00, '2025-02-01', 'VRN002', 'UID1002', 'HDB005'),  
( 'UTR003', '2025-01-01', 80.00, '2025-02-01', 'VRN003', 'UID1003', 'HDB020'),
```

```
(('UTR004','2025-02-01', 80.00,'2025-03-01','VRN004','UID1004','HDB003'),
('UTR005','2025-02-01',100.00,'2025-03-01','VRN005','UID1005','HDB005'),
('UTR006','2025-02-01', 80.00,'2025-03-01','VRN006','UID1006','HDB016'),
('UTR007','2025-02-01', 30.00,'2025-03-01','VRN007','UID1007','HDB003'),
('UTR008','2025-03-01',100.00,'2025-04-01','VRN008','UID1008','HDB001'),
('UTR009','2025-03-01', 80.00,'2025-04-01','VRN009','UID1009','HDB013'),
('UTR010','2025-03-01', 80.00,'2025-04-01','VRN010','UID1010','HDB002'),
('UTR011','2025-03-01', 50.00,'2025-04-01','VRN011','UID1011','HDB014'),
('UTR012','2025-03-01', 30.00,'2025-04-01','VRN012','UID1012','HDB004'),
('UTR013','2025-03-01', 30.00,'2025-04-01','VRN013','UID1013','HDB015'),
('UTR014','2025-04-01',190.00,'2025-05-01','VRN014','UID1014','HDB006'),
('UTR015','2025-04-01',170.00,'2025-05-01','VRN015','UID1015','HDB017'),
('UTR016','2025-04-01',190.00,'2025-05-01','VRN016','UID1016','HDB007'),
('UTR017','2025-04-01',100.00,'2025-05-01','VRN017','UID1017','HDB019'),
('UTR018','2025-05-01',100.00,'2025-06-01','VRN018','UID1018','HDB008'),
('UTR019','2025-05-01', 30.00,'2025-06-01','VRN019','UID1019','HDB020'),
('UTR020','2025-05-01', 560.00,'2025-12-01','VRN020','UID1020','HDB010'),
('UTR021','2025-02-01', 1040.00,'2026-03-01','VRN004','UID1021','HDB010'),
('UTR022','2025-02-01', 800.00,'2025-12-01','VRN007','UID1022','HDB010'),
('UTR023','2025-05-01', 560.00,'2025-12-01','VRN022','UID1023','HDB010'),
('UTR024','2025-05-01', 560.00,'2025-12-01','VRN023','UID1024','HDB010'),
('UTR025','2025-05-01', 560.00,'2025-12-01','VRN025','UID1025','HDB010'),
('UTR026','2025-05-01', 850.00,'2025-12-01','VRN021','UID1026','HDB010'),
('UTR027','2025-05-01', 640.00,'2026-01-01','VRN024','UID1027','HDB010'),
('UTR028','2025-05-01', 560.00,'2025-12-01','VRN007','UID1028','HDB010'),
('UTR029','2025-05-01', 1020.00,'2026-01-01','VRN016','UID1029','HDB010'),
('UTR030','2025-05-01', 560.00,'2025-12-01','VRN010','UID1030','HDB010'),
('UTR031','2025-05-01', 560.00,'2025-12-01','VRN009','UID1031','HDB010');
```

--(UTR 4,7,21, 22 is for Q6)

```
-----

CREATE TABLE ParkingRecord (
entry_datetime DATETIME NOT NULL,
VRN CHAR(6) NOT NULL,
carpark_ID CHAR(6) NOT NULL,
UID VARCHAR(10) NOT NULL,
exit_datetime DATETIME,

PRIMARY KEY (entry_datetime, VRN, carpark_ID),
```

UNIQUE (UID, entry_datetime, exit_datetime),

FOREIGN KEY (VRN) REFERENCES Vehicle1(VRN)

ON DELETE CASCADE,

FOREIGN KEY (carpark_ID) REFERENCES NormalCarpark(ID)

ON DELETE CASCADE,

FOREIGN KEY (UID) REFERENCES ParkingModeRecord(UID)

ON DELETE CASCADE,

);

INSERT INTO ParkingRecord(entry_datetime, VRN, carpark_ID, UID, exit_datetime)
VALUES

('2025-11-05 07:00:00','VRN008','HDB005','UID0001','2025-11-05 17:00:00'),
('2025-11-04 19:00:00','VRN010','HDB003','UID0002','2025-11-05 03:00:00'),
('2025-11-03 07:00:00','VRN004','HDB010','UID0003','2025-11-03 11:00:00'),
('2025-11-02 07:00:00','VRN012','HDB009','UID0004','2025-11-02 07:00:00'),
('2025-11-01 07:00:00','VRN002','HDB019','UID0005','2025-11-01 17:10:00'),
('2025-11-06 17:00:00','VRN014','HDB019','UID0006','2025-11-06 23:00:00'),
('2025-11-07 07:00:00','VRN016','HDB018','UID0007','2025-11-07 08:00:00'),
('2025-11-02 07:00:00','VRN018','HDB019','UID0008','2025-11-02 22:30:00'),
('2025-11-03 17:00:00','VRN003','HDB019','UID0009','2025-11-03 19:00:00'),
('2025-11-05 07:00:00','VRN010','HDB004','UID0010','2025-11-05 11:00:00'),
('2025-11-04 07:00:00','VRN001','HDB019','UID0011','2025-11-04 21:00:00'),
('2025-11-02 07:00:00','VRN009','HDB019','UID0012','2025-11-02 22:30:00'),
('2025-11-01 17:00:00','VRN011','HDB016','UID0013','2025-11-01 19:00:00'),
('2025-11-06 07:00:00','VRN013','HDB018','UID0014','2025-11-06 09:30:00'),
('2025-11-05 07:00:00','VRN006','HDB009','UID0015','2025-11-05 13:00:00'),
('2025-11-04 07:00:00','VRN015','HDB019','UID0016','2025-11-04 15:00:00'),
('2025-11-05 17:00:00','VRN017','HDB018','UID0017','2025-11-05 20:00:00'),
('2025-11-03 07:00:00','VRN018','HDB017','UID0018','2025-11-03 19:00:00'),
('2025-11-01 17:00:00','VRN019','HDB016','UID0019','2025-11-02 02:12:00'),
('2025-11-05 07:00:00','VRN020','HDB019','UID0020','2025-11-05 11:00:00'),
('2025-01-05 08:00:00','VRN001','HDB006','UID1001','2025-01-05 18:00:00'),
('2025-01-15 09:30:00','VRN002','HDB005','UID1002','2025-01-15 17:30:00'),
('2025-01-20 07:45:00','VRN003','HDB020','UID1003','2025-01-20 19:00:00'),
('2025-02-05 08:15:00','VRN004','HDB003','UID1004','2025-02-05 18:15:00'),
('2025-02-10 09:00:00','VRN005','HDB005','UID1005','2025-02-10 17:45:00'),
('2025-02-12 07:30:00','VRN006','HDB016','UID1006','2025-02-12 19:15:00'),
('2025-02-20 08:00:00','VRN007','HDB003','UID1007','2025-02-20 18:00:00'),
('2025-03-05 08:30:00','VRN008','HDB001','UID1008','2025-03-05 18:30:00'),
('2025-03-10 09:00:00','VRN009','HDB013','UID1009','2025-03-10 18:00:00'),

```
('2025-03-12 08:00:00','VRN010','HDB002','UID1010','2025-03-12 17:30:00'),
('2025-03-15 07:45:00','VRN011','HDB014','UID1011','2025-03-15 18:15:00'),
('2025-03-20 08:30:00','VRN012','HDB004','UID1012','2025-03-20 18:00:00'),
('2025-03-22 08:15:00','VRN013','HDB015','UID1013','2025-03-22 18:30:00'),
('2025-04-05 08:00:00','VRN014','HDB006','UID1014','2025-04-05 18:00:00'),
('2025-04-10 09:00:00','VRN015','HDB017','UID1015','2025-04-10 18:00:00'),
('2025-04-12 07:45:00','VRN016','HDB007','UID1016','2025-04-12 18:15:00'),
('2025-04-15 08:30:00','VRN017','HDB019','UID1017','2025-04-15 18:30:00'),
('2025-05-05 08:00:00','VRN018','HDB008','UID1018','2025-05-05 18:00:00'),
('2025-05-10 09:15:00','VRN019','HDB020','UID1019','2025-05-10 18:45:00'),
('2025-05-12 08:00:00','VRN020','HDB010','UID1020','2025-05-12 17:30:00'),
('2025-01-12 07:45:00','VRN007','HDB011','UID0021','2025-04-12 18:15:00'),
('2025-02-12 07:45:00','VRN007','HDB012','UID0022','2025-04-12 18:15:00'),
('2025-03-12 07:45:00','VRN007','HDB013','UID0023','2025-04-12 18:15:00'),
('2025-04-12 07:45:00','VRN010','HDB014','UID1021','2025-04-12 18:15:00'),
('2025-03-06 08:30:00','VRN010','HDB015','UID1022','2025-03-05 18:30:00'),
('2025-03-05 08:30:00','VRN010','HDB016','UID1023','2025-03-05 18:30:00');
```

```
-----
CREATE TABLE Tx_ShortTermParking (
  UID VARCHAR(10) NOT NULL,
  entry_datetime DATETIME NOT NULL,
  exit_datetime DATETIME NOT NULL,
  total_amount_paid FLOAT NOT NULL CHECK (total_amount_paid >= 0),
  NRIC CHAR(9) NOT NULL,
  vehicle_type VARCHAR(20) NOT NULL,

  PRIMARY KEY (UID, entry_datetime, exit_datetime),

  FOREIGN KEY (NRIC) REFERENCES Owner(NRIC)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

  FOREIGN KEY (UID, entry_datetime, exit_datetime)
    REFERENCES ParkingRecord(UID, entry_datetime, exit_datetime)
    ON DELETE NO ACTION,

  FOREIGN KEY (UID)
    REFERENCES ParkingModeRecord(UID)
    ON DELETE CASCADE
);
-----
```

```

INSERT INTO Tx_ShortTermParking (UID, entry_datetime, exit_datetime,
total_amount_paid, NRIC, vehicle_type)
VALUES
('UID0001','2025-11-05 07:00:00','2025-11-05 17:00:00', 8.4, 'M0000001A', 'Car'),
('UID0002','2025-11-04 19:00:00','2025-11-05 03:00:00', 4.8, 'M0000002B', 'Car'),
('UID0003','2025-11-03 07:00:00','2025-11-03 11:00:00', 2.4, 'M0000003C',
'Motorcycle'),
('UID0004','2025-11-02 07:00:00','2025-11-02 07:00:00', 0, 'M0000004D', 'Motorcycle'),
('UID0005','2025-11-01 07:00:00','2025-11-01 17:10:00', 12.2, 'M0000005E', 'Car'),
('UID0006','2025-11-06 17:00:00','2025-11-06 23:00:00', 3.6, 'M0000006F',
'Commercial'),
('UID0007','2025-11-07 07:00:00','2025-11-07 08:00:00', 1.2, 'M0000007G',
'Commercial'),
('UID0008','2025-11-02 07:00:00','2025-11-02 22:30:00', 0, 'M0000008H', 'Car'),
('UID0009','2025-11-03 17:00:00','2025-11-03 19:00:00', 1.2, 'M0000009I', 'Car'),
('UID0010','2025-11-05 07:00:00','2025-11-05 11:00:00', 4.8, 'M0000010J', 'Car'),
('UID0011','2025-11-04 07:00:00','2025-11-04 21:00:00', 16.8, 'F0000001A', 'Car'),
('UID0012','2025-11-02 07:00:00','2025-11-02 22:30:00', 0, 'F0000002B', 'Car'),
('UID0013','2025-11-01 17:00:00','2025-11-01 19:00:00', 0.6, 'F0000003C',
'Motorcycle'),
('UID0014','2025-11-06 07:00:00','2025-11-06 09:30:00', 1.5, 'F0000004D',
'Motorcycle'),
('UID0015','2025-11-05 07:00:00','2025-11-05 13:00:00', 7.2, 'F0000005E', 'Car'),
('UID0016','2025-11-04 07:00:00','2025-11-04 15:00:00', 9.6, 'F0000006F',
'Commercial'),
('UID0017','2025-11-05 17:00:00','2025-11-05 20:00:00', 1.8, 'F0000007G', 'Car'),
('UID0018','2025-11-03 07:00:00','2025-11-03 19:00:00', 14.4, 'F0000008H', 'Car'),
('UID0019','2025-11-01 17:00:00','2025-11-02 02:12:00', 11.0, 'F0000009I',
'Commercial'),
('UID0020','2025-11-05 07:00:00','2025-11-05 11:00:00', 4.8, 'F0000010J', 'Car');

```

```

CREATE TABLE ShortTermRate(
    day VARCHAR(4),
    time VARCHAR(30),
    vehicle_type VARCHAR(20),
    parking_rate FLOAT NOT NULL CHECK( parking_rate >= 0 ),

    PRIMARY KEY (day, time, vehicle_type)
);

```

```

-----
INSERT INTO ShortTermRate( day, time, vehicle_type, parking_rate )

```

VALUES

('Mon', '7.00am to 5.00pm', 'Car', 1.2),
('Mon', '5.00pm to 7.00am', 'Car', 0.6),
('Mon', '7.00am to 5.00pm', 'Commercial', 1.2),
('Mon', '5.00pm to 7.00am', 'Commercial', 0.6),
('Mon', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Mon', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Tue', '7.00am to 5.00pm', 'Car', 1.2),
('Tue', '5.00pm to 7.00am', 'Car', 0.6),
('Tue', '7.00am to 5.00pm', 'Commercial', 1.2),
('Tue', '5.00pm to 7.00am', 'Commercial', 0.6),
('Tue', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Tue', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Wed', '7.00am to 5.00pm', 'Car', 1.2),
('Wed', '5.00pm to 7.00am', 'Car', 0.6),
('Wed', '7.00am to 5.00pm', 'Commercial', 1.2),
('Wed', '5.00pm to 7.00am', 'Commercial', 0.6),
('Wed', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Wed', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Thu', '7.00am to 5.00pm', 'Car', 1.2),
('Thu', '5.00pm to 7.00am', 'Car', 0.6),
('Thu', '7.00am to 5.00pm', 'Commercial', 1.2),
('Thu', '5.00pm to 7.00am', 'Commercial', 0.6),
('Thu', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Thu', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Fri', '7.00am to 5.00pm', 'Car', 1.2),
('Fri', '5.00pm to 7.00am', 'Car', 0.6),
('Fri', '7.00am to 5.00pm', 'Commercial', 1.2),
('Fri', '5.00pm to 7.00am', 'Commercial', 0.6),
('Fri', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Fri', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Sat', '7.00am to 5.00pm', 'Car', 1.2),
('Sat', '5.00pm to 7.00am', 'Car', 0.6),
('Sat', '7.00am to 5.00pm', 'Commercial', 1.2),
('Sat', '5.00pm to 7.00am', 'Commercial', 0.6),

```

('Sat', '7.00am to 5.00pm', 'Motorcycle', 0.6),
('Sat', '5.00pm to 7.00am', 'Motorcycle', 0.3),

('Sun', '7.00am to 10.30pm', 'Car', 0),
('Sun', '10.30pm to 7.00am', 'Car', 0.6),
('Sun', '7.00am to 10.30pm', 'Commercial', 0),
('Sun', '10.30pm to 7.00am', 'Commercial', 0.6),
('Sun', '7.00am to 10.30pm', 'Motorcycle', 0),
('Sun', '10.30pm to 7.00am', 'Motorcycle', 0.3);

```

```

CREATE TABLE Rules (
rule_no CHAR(6),
description VARCHAR(100),
vehicle_type VARCHAR(20),
action_fine INT ,

```

```

PRIMARY KEY(rule_no)
);

```

```

INSERT INTO Rules (rule_no, description, vehicle_type, action_fine) VALUES
('R00003', 'parking in a place other than in a parking lot', 'Motorcycle', 35),
('R00006', 'parking in a place other than in a parking lot', 'Car', 90),
('R00010', 'parking in a parking lot not designated for the use of that vehicle', 'Car', 70),
('R00018', 'parking in a season parking place without valid season parking', 'Car', 70),
('R00007', 'failure to pay parking charge before leaving a parking place', 'Car', 70),
('R00001', 'parking without displaying an authorised label', 'Car',200),
('R00002', 'parking without displaying an authorised label', 'Motorcycle', 100),
('R00012', 'failure to obey signs exhibited in the parking place', 'Commercial', 100),
('R00013', 'parking beyond the boundaries of the parking lot thereby causing
obstruction', 'Car', 70),
('R00004', 'using a parking place for unauthorised purpose', 'Car', 80),
('R00005', 'using a parking place for unauthorised purpose', 'Motorcycle', 80),
('R00008', 'parking against traffic flow', 'Car',70);

```

```

CREATE TABLE Offence(
OffenceID CHAR(6),
entry_datetime DATETIME,
VRN CHAR(6),
carpark_identification_number CHAR(6),

```

```
offence_datetime DATETIME,  
rule_no CHAR(6),
```

```
PRIMARY KEY(OffenceID, entry_datetime, VRN, carpark_identification_number),
```

```
FOREIGN KEY(rule_no) REFERENCES Rules(rule_no)
```

```
ON DELETE SET NULL ON UPDATE CASCADE,
```

```
FOREIGN KEY (entry_datetime, VRN, carpark_identification_number) REFERENCES
```

```
ParkingRecord(entry_datetime, VRN, carpark_ID)
```

```
ON DELETE CASCADE ON UPDATE CASCADE
```

```
);
```

```
INSERT INTO Offence(OffenceID, entry_datetime, VRN, carpark_identification_number,  
offence_datetime, rule_no) VALUES
```

```
('Of0001', '2025-01-05 08:00:00', 'VRN001', 'HDB006', '2025-01-05 09:45:00', 'R00010'),  
( 'Of0002', '2025-01-20 07:45:00', 'VRN003', 'HDB020', '2025-01-20 09:10:00', 'R00013'),  
( 'Of0003', '2025-02-10 09:00:00', 'VRN005', 'HDB005', '2025-02-10 15:15:00', 'R00001'),  
( 'Of0004', '2025-02-20 08:00:00', 'VRN007', 'HDB003', '2025-02-20 09:15:00', 'R00005'),  
( 'Of0005', '2025-01-15 09:30:00', 'VRN002', 'HDB005', '2025-01-15 13:15:00', 'R00008'),  
( 'Of0006', '2025-03-15 07:45:00', 'VRN011', 'HDB014', '2025-03-15 09:25:00', 'R00003'),  
( 'Of0007', '2025-03-12 08:00:00', 'VRN010', 'HDB002', '2025-03-12 10:15:00', 'R00010'),  
( 'Of0008', '2025-03-20 08:30:00', 'VRN012', 'HDB004', '2025-03-20 09:15:00', 'R00002'),  
( 'Of0009', '2025-01-15 09:30:00', 'VRN002', 'HDB005', '2025-01-15 15:15:00', 'R00018'),  
( 'Of0010', '2025-02-05 08:15:00', 'VRN004', 'HDB003', '2025-02-05 09:15:00', 'R00004'),  
( 'Of0011', '2025-03-22 08:15:00', 'VRN013', 'HDB015', '2025-03-22 09:15:00', 'R00002'),  
( 'Of0012', '2025-04-10 09:00:00', 'VRN015', 'HDB017', '2025-04-10 09:45:00', 'R00012'),  
( 'Of0013', '2025-02-10 09:00:00', 'VRN005', 'HDB005', '2025-02-10 10:15:00', 'R00008'),  
( 'Of0014', '2025-02-20 08:00:00', 'VRN007', 'HDB003', '2025-02-20 09:15:00', 'R00002'),  
( 'Of0015', '2025-02-12 07:30:00', 'VRN006', 'HDB016', '2025-02-12 09:15:00', 'R00006'),  
( 'Of0016', '2025-05-10 09:15:00', 'VRN019', 'HDB020', '2025-05-10 11:35:00', 'R00003'),  
( 'Of0017', '2025-04-12 07:45:00', 'VRN016', 'HDB007', '2025-04-12 10:55:00', 'R00002'),  
( 'Of0018', '2025-03-12 08:00:00', 'VRN010', 'HDB002', '2025-03-12 09:15:00', 'R00004'),  
( 'Of0019', '2025-03-05 08:30:00', 'VRN008', 'HDB001', '2025-03-05 11:23:00', 'R00018'),  
( 'Of0020', '2025-04-12 07:45:00', 'VRN016', 'HDB007', '2025-04-12 09:35:00', 'R00012'),  
( 'Of0021', '2025-01-12 07:45:00', 'VRN007', 'HDB011', '2025-04-12 09:35:00', 'R00012'),  
( 'Of0022', '2025-02-12 07:45:00', 'VRN007', 'HDB012', '2025-04-12 09:35:00', 'R00002'),  
( 'Of0023', '2025-03-12 07:45:00', 'VRN007', 'HDB013', '2025-04-12 09:35:00', 'R00008'),  
( 'Of0024', '2025-04-12 07:45:00', 'VRN010', 'HDB014', '2025-04-12 09:35:00', 'R00005'),  
( 'Of0025', '2025-03-06 08:30:00', 'VRN010', 'HDB015', '2025-03-05 11:23:00', 'R00004'),  
( 'Of0026', '2025-03-05 08:30:00', 'VRN010', 'HDB016', '2025-03-05 11:23:00', 'R00012');
```

```
-----
```

Queries to solve Appendix B

1. For each MSCP, count how many vehicles park for less than an hour during weekdays (Mon – Fri). Breakdown the records by vehicle types (i.e. car, motorcycle, and commercial vehicle).

```
WITH Vehicle AS
(SELECT t.vehicle_type, v.VRN
FROM VehicleModel t
JOIN Vehicle1 v
ON t.model = v.model)
SELECT c.ID, v.vehicle_type, COUNT(p.VRN) AS number
FROM MSCP c
JOIN ParkingRecord p
ON c.ID = p.carpark_ID
AND DATEPART(WEEKDAY, p.entry_datetime) NOT IN (5,6)
AND DATEDIFF(hour, p.exit_datetime, p.entry_datetime) < 1
JOIN Vehicle v
ON v.VRN = p.VRN
GROUP BY c.ID, v.vehicle_type;
```

	ID	vehicle_type	number
1	HDB001	Car	1
2	HDB005	Car	3
3	HDB006	Car	1
4	HDB008	Car	1
5	HDB009	Car	1
6	HDB014	Car	1
7	HDB018	Car	1
8	HDB019	Car	7
9	HDB006	Commercial	1
10	HDB007	Commercial	1
11	HDB019	Commercial	1
12	HDB009	Motorcycle	1
13	HDB011	Motorcycle	1
14	HDB014	Motorcycle	1

2. For each carpark, list down the total revenue collected for both short term parking as season parking separately.

--Season Parking

```
SELECT carpark_ID, sum(amount_paid) as total_season_paid
FROM r_transaction_seasonpass_1
GROUP BY carpark_ID;
```

--Short term parking

```
SELECT n.carpark_ID, round(sum(t.total_amount_paid),2) as short_term_total
FROM parkingrecord as n
JOIN tx_shorttermparking as t
```

```

ON n.UID = t.UID
GROUP BY n.carpark_ID;
--Display both revenues in one result table
WITH season AS (
    SELECT carpark_ID, SUM(amount_paid) AS total_season_paid
    FROM r_transaction_seasonpass_1
    GROUP BY carpark_ID
),
short_term AS (
    SELECT n.carpark_ID, ROUND(SUM(t.total_amount_paid), 2) AS short_term_total
    FROM parkingrecord AS n
    JOIN tx_shorttermparking AS t ON n.UID = t.UID
    GROUP BY n.carpark_ID
)
SELECT
    N.ID,
    COALESCE(s.total_season_paid, 0) AS total_season_paid,
    COALESCE(st.short_term_total, 0) AS short_term_total
FROM NormalCarpark AS N
LEFT OUTER JOIN season AS s ON N.ID = s.carpark_ID
LEFT OUTER JOIN short_term AS st ON N.ID = st.carpark_ID;

```

	ID	total_season_paid	short_term_total
1	HDB001	100	0
2	HDB002	80	0
3	HDB003	110	4.8
4	HDB004	30	4.8
5	HDB005	200	8.4
6	HDB006	290	0
7	HDB007	190	0
8	HDB008	100	0
9	HDB009	0	7.2
10	HDB010	8270	2.4
11	HDB011	0	0
12	HDB012	0	0
13	HDB013	80	0
14	HDB014	50	0
15	HDB015	30	0
16	HDB016	80	11.6
17	HDB017	170	14.4
18	HDB018	0	4.5
19	HDB019	100	48.2
20	HDB020	110	0

3. List down the season parking holder who are not resident at the HDB block that linked to the carpark. For each pass holder, count how many months they have subscribe to the season parking, break down by different carpark

```
SELECT o.NRIC, o.name, r.carpark_ID, SUM(
    DATEDIFF(
        MONTH,
        r.date_time,
        CASE
            WHEN r.expiry_date > GETDATE() THEN GETDATE()
            ELSE r.expiry_date
        END
    ) + 1
) AS total_months_subscribed
FROM
R_Transaction_SeasonPass_1 AS r
JOIN Vehicle1 AS v ON r.VRN = v.VRN
JOIN Owner AS o ON v.NRIC = o.NRIC
WHERE o.postal_code NOT IN (SELECT b.postal_code
FROM BlockAddress AS b
WHERE r.carpark_ID = b.carpark_ID)
GROUP BY o.NRIC, o.name, r.carpark_ID
```

	NRIC	name	carpark_ID	total_months_subscribed
1	M0000001A	John	HDB005	7
2	F0000002B	Angeline	HDB010	7
3	F0000010J	Minnie	HDB010	7
4	M0000002B	Peter	HDB010	7
5	M0000003C	Jack	HDB010	27
6	M0000007G	Jack	HDB010	7

4. List down the name of vehicle owner and their vehicle registration number who committed parking offences at the carpark where they are not the residents. Count the number of offences and total amount of fines

```
SELECT w.name, x.VRN, count(*) AS offence_count, SUM(z.action_fine) as total_fine
FROM Vehicle1 x, Offence y, Rules z, Owner w, BlockAddress k
WHERE x.VRN = y.VRN
AND y.rule_no = z.rule_no
AND w.NRIC = x.NRIC
AND w.postal_code = k.postal_code
AND y.carpark_identification_number <> k.carpark_id
GROUP BY w.name, x.VRN;
```

	name	VRN	offence_count	total_fine
1	Jack	VRN007	3	270
2	Peter	VRN010	3	260

5. When 90% of season parking passes for a particular month and carpark (all parking lots) are sold, only the residents living in the HDB block that linked to the carpark who have not purchased any season parking pass can purchase the remaining pass. Prepare your data and query to simulate the success and fail scenario for non-resident.

--Separate Files

--Question 5 Trigger Initialization

--Create Trigger for Simulation

CREATE TRIGGER BuySimulation

ON R_Transaction_SeasonPass_1

INSTEAD OF INSERT

AS

BEGIN

IF EXISTS (

SELECT 1

FROM inserted i

WHERE

(CAST(

(SELECT COUNT(*)

FROM R_Transaction_SeasonPass_1 r

WHERE r.carpark_ID = i.carpark_ID

AND r.expiry_date >= GETDATE())

AS FLOAT

)/

(SELECT SUM(number_of_lots)

FROM ParkingLots

WHERE ID = i.carpark_ID AND color = 'Red') >= 0.9)

AND

((SELECT b.carpark_id

FROM Vehicle1 v

JOIN Owner o ON v.NRIC = o.NRIC

JOIN BlockAddress b ON o.postal_code = b.postal_code

WHERE v.VRN = i.VRN) <> i.carpark_ID)

)

BEGIN

RAISERROR ('Purchase denied: 90%% red lots filled and buyer not resident.', 16, 1);

```

RETURN;
END;

IF EXISTS (
SELECT 1
FROM inserted i
WHERE
(CAST(
(SELECT COUNT(*)
FROM R_Transaction_SeasonPass_1 r
WHERE r.carpark_ID = i.carpark_ID
AND r.expiry_date >= GETDATE())
AS FLOAT
)/
(SELECT SUM(number_of_lots)
FROM ParkingLots
WHERE ID = i.carpark_ID AND color = 'Red') >= 1)
)
BEGIN
RAISERROR ('Purchase denied: All pass has been sold', 16, 1);
RETURN;
END;

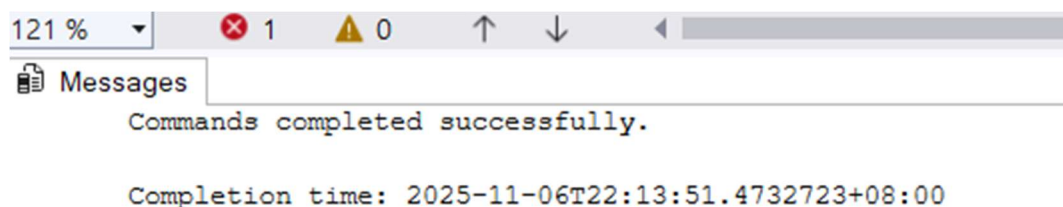
```

```

INSERT INTO R_Transaction_SeasonPass_1 (UTR, date_time, amount_paid,
expiry_date, VRN, UID, carpark_ID)
SELECT UTR, date_time, amount_paid, expiry_date, VRN, UID, carpark_ID
FROM inserted;
PRINT 'Purchase Successful!';
END;

```

--The Queries below must be called from a separate files as SQL Server does not allow trigger initialization and normal queries in one file



--Question 5 Simulation

--Initial status of R_Transaction_SeasonPass_1

```
SELECT *  
FROM R_Transaction_SeasonPass_1;
```

--Begin Simulation

--Non-resident trying to buy limited season pass at HDB010

```
INSERT INTO R_Transaction_SeasonPass_1  
(UTR, date_time, amount_paid, expiry_date, VRN, UID, carpark_ID)  
VALUES  
('UTR505','2025-05-01',700.00,'2025-12-01','VRN008','UID1505','HDB010');
```

--Resident trying to buy limited season pass

```
INSERT INTO R_Transaction_SeasonPass_1  
(UTR, date_time, amount_paid, expiry_date, VRN, UID, carpark_ID)  
VALUES  
('UTR506','2025-05-01',700.00,'2025-12-01','VRN023','UID1506','HDB010');
```

--Non-resident trying to buy normal season pass (non-limited) at HDB005

```
INSERT INTO R_Transaction_SeasonPass_1  
(UTR, date_time, amount_paid, expiry_date, VRN, UID, carpark_ID)  
VALUES  
('UTR507','2025-05-01',700.00,'2025-12-01','VRN008','UID1507','HDB005');
```

--Resident trying to buy pass when there is none left

```
INSERT INTO R_Transaction_SeasonPass_1  
(UTR, date_time, amount_paid, expiry_date, VRN, UID, carpark_ID)  
VALUES  
('UTR508','2025-05-01',700.00,'2025-12-01','VRN025','UID1508','HDB010');
```

--Confirm status of R_Transaction_SeasonPass_1

```
SELECT *  
FROM R_Transaction_SeasonPass_1;
```

```

(31 rows affected)
Msg 50000, Level 16, State 1, Procedure BuySimulation, Line 31 [Batch Start Line 0]
Purchase denied: 90% red lots filled and buyer not resident.

(1 row affected)

(1 row affected)
Purchase Successful!

(1 row affected)

(1 row affected)
Purchase Successful!

(1 row affected)
Msg 50000, Level 16, State 1, Procedure BuySimulation, Line 51 [Batch Start Line 0]
Purchase denied: All pass has been sold

```

Results		Messages					
24	UTR024	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN023	UID1024	HDB010
25	UTR025	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN025	UID1025	HDB010
26	UTR026	2025-05-01 00:00:00.000	850	2025-12-01 00:00:00.000	VRN021	UID1026	HDB010
27	UTR027	2025-05-01 00:00:00.000	640	2026-01-01 00:00:00.000	VRN024	UID1027	HDB010
28	UTR028	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN007	UID1028	HDB010
29	UTR029	2025-05-01 00:00:00.000	1020	2026-01-01 00:00:00.000	VRN016	UID1029	HDB010
30	UTR030	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN010	UID1030	HDB010
31	UTR031	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN009	UID1031	HDB010
	UTR	date_time	amount_paid	expiry_date	VRN	UID	carpark_ID
26	UTR026	2025-05-01 00:00:00.000	850	2025-12-01 00:00:00.000	VRN021	UID1026	HDB010
27	UTR027	2025-05-01 00:00:00.000	640	2026-01-01 00:00:00.000	VRN024	UID1027	HDB010
28	UTR028	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN007	UID1028	HDB010
29	UTR029	2025-05-01 00:00:00.000	1020	2026-01-01 00:00:00.000	VRN016	UID1029	HDB010
30	UTR030	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN010	UID1030	HDB010
31	UTR031	2025-05-01 00:00:00.000	560	2025-12-01 00:00:00.000	VRN009	UID1031	HDB010
32	UTR506	2025-05-01 00:00:00.000	700	2025-12-01 00:00:00.000	VRN023	UID1506	HDB010
33	UTR507	2025-05-01 00:00:00.000	700	2025-12-01 00:00:00.000	VRN008	UID1507	HDB005

6. Identify the household (address) that purchases more than three season parking for the same month. List down the name of vehicle owners and the vehicle registration number

/*

1. We first identify the household address (residential unit here) that purchase more than 3 times in same month, make it a temporary view table, R2.

2. After that, we filter out the tuple that fulfill this condition, which mean the selected tuple in R2 .

3. Then we select the distinct VRN, owner names , and their full address atributes.

*/

WITH R2 AS

(SELECT O.postal_code, O.unit_number, FORMAT(T.date_time, 'yyyy-MM') as month_key

```

FROM R_Transaction_SeasonPass_1 T
JOIN Vehicle1 V ON T.VRN = V.VRN
JOIN Owner O ON O.NRIC = V.NRIC
GROUP BY O.postal_code, O.unit_number, FORMAT(T.date_time, 'yyyy-MM')
HAVING COUNT(*) > 3 )
SELECT DISTINCT O.name, T.VRN, O.postal_code, B.hdb_block_number, B.street_name,
O.unit_number
FROM R_Transaction_SeasonPass_1 T
JOIN Vehicle1 V ON T.VRN = V.VRN
JOIN Owner O ON O.NRIC = V.NRIC
JOIN BlockAddress B ON B.postal_code = O.postal_code
JOIN R2
    ON R2.postal_code = O.postal_code
    AND R2.unit_number = O.unit_number
    AND R2.month_key = FORMAT(T.date_time, 'yyyy-MM');

```

-- This will print the full address of the person, if address no need print out, just delete the relevant selected attributes and no need join the hdb address.

	name	VRN	postal_code	hdb_block_number	street_name	unit_number
1	Jack	VRN004	310125	25A	Toa Payoh Lor 2	#03-103
2	Jack	VRN007	310125	25A	Toa Payoh Lor 2	#03-103